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Per- and polyfluoroalkyl substances (PFAS) in paired tap water and house dust from United States homes

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Disclaimer

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Ethical approval

EPA's Office of Research and Development (ORD), Center for Public Health and Environmental Assessment was not directly engaged in the collection of information from human subjects. HUD's contractor, QuanTech, conducted the field study and collected tap water and house dust samples. QuanTech received IRB Approval CR00077983 for HUD OHHLHC - AHHS II, American Healthy Homes Survey (AHHS) II (Pro00019737). According to the requirements of EPA Order 1000.17 A (Policy and Procedures on Protection of Human Research Subjects) and EPA Regulation 40 CFR 26 (Protection of Human Subjects), it was determined that the EPA investigators were not engaged in human subjects research (HSR-001225).

CRediT authorship contribution statement

David Cox: Writing – review & editing, Investigation, Data curation. **Karen Bradham:** Writing – review & editing, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Eugene Pinzer:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Warren Friedman:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Gary DeWalt:** Writing – review & editing, Investigation, Data curation. **Peter Ashley:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Jeffrey M. Minucci:** Writing – review & editing, Formal analysis, Conceptualization. **Christopher Fuller:** Writing – review & editing, Methodology, Data curation. **Kelsey E. Miller:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Jason Boettger:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Elaine A. Cohen Hubal:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Nicole M. DeLuca:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **James McCord:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indenv.2024.100033.

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Abstract

Most people in the United States have been exposed to per- and polyfluoroalkyl substances (PFAS) which have been linked to a wide array of adverse health conditions in adults and children. The consumption of contaminated drinking water is an important human exposure pathway to PFAS. Residential sources also contribute to PFAS exposure through dermal contact and ingestion of house dust, which acts as an aggregate of chemicals from sources like furnishing materials and consumer products. The U.S. Department of Housing and Urban Development (HUD) conducted the first nationwide survey of residential hazards called the American Healthy Homes Survey (AHHS) in 2005, followed by a second survey (AHHS II) in 2017. The U.S. Environmental Protection Agency (EPA) collaborated with HUD on both efforts and subsequently analyzed PFAS in household tap water and house dust collected from the same homes during the AHHS II study. This study leverages these paired samples to investigate potentially important exposure sources and pathways in the residential environment. Here we report results for paired household tap water and house dust samples from 241 homes for 13 and 16 PFAS chemicals, respectively. All 13 targeted chemicals were detected in the household tap water samples with detections ranging from 100 % for PFBS to 1 % for PFNS, and all 16 targeted chemicals were detected in the house dust samples with detections ranging from 97 % for PFOA to 9 % for PFNS. Four chemicals (PFOA, PFOS, PFHxS, and PFHpA) were measured above the limit of detection in at least 50 % of the samples in both media. All households had at least one of the targeted PFAS detected in both their tap water and house dust. Results provided evidence that geographical factors, such as proximity to ambient contamination sources, were main drivers of PFAS contamination in tap water, while PFAS contamination in house dust was driven mainly by within-home sources. Exposure estimates calculated from the measured PFAS concentrations highlight the importance of addressing potential sources of exposure to PFAS within homes in addition to ambient sources affecting communities' drinking water, particularly to reduce children's exposure to these chemicals.

Keywords

Indoor environment; PFAS sources; Healthy homes; Drinking water; Human exposure

1. Introduction

Biomonitoring data from the National Health and Nutrition Examination Survey (NHANES) has shown that most people in the United States have been exposed to per- and polyfluoroalkyl substances (PFAS), a group of thousands of synthetic chemicals used since the 1940s for their non-stick, stain- and water-repellant, and fire suppressing properties [11,24,38,71,82,83,42,76]. Exposure to PFAS has been linked to adverse conditions regarding reproductive health, child development, cancers, cholesterol levels, endocrine system function, liver and kidney disease, and immune response [20,71]. Mitigating human exposure to PFAS is difficult due to their persistent nature and near ubiquitous presence in the natural and built environments. Because some PFAS can remain in the human body for years, the effects of these chemicals can occur long after initial exposure [98,43,63]. Beginning in the early 2000s, a voluntary phase out of several long-chain "legacy" PFAS

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was initiated by U.S. PFAS manufacturers due to growing awareness of their adverse health effects. However, manufacturing has largely shifted to shorter-chain substitutes and alternate fluorinated chemistry [58].

The consumption of contaminated drinking water is a main human exposure pathway to PFAS and can be the cause of accumulative exposures over long periods of time before it is identified [71]. Thus, the U.S. Environmental Protection Agency (EPA) has recently proposed national primary drinking water regulations for 6 PFAS to protect public health [84]. The Third Unregulated Contaminant Monitoring Rule (UCMR3) program helped to identify large public water systems with PFAS-contaminated drinking water, and some researchers have suggested that UCMR3 data are correlated with reported ambient sources of PFAS contamination [31]. Additional data on the occurrence of PFAS in public water systems are currently being collected under the Fifth Unregulated Contaminant Monitoring Rule (UCMR5), which will provide baseline monitoring data to support the proposed regulations [85]. A presumptive contamination approach was also proposed to identify unsampled areas with the potential for PFAS contamination in the environment based on the types of industry, waste, and past or current aqueous film-forming foam (AFFF) use [67]. However, nationwide characterizations and investigations of the drivers of PFAS levels in drinking water from both public and private sources of household tap water, particularly at the point-of-use, are currently still limited [69].

Where PFAS in tap water have been found to be significant predictors of PFAS levels in plasma, estimates of contributions of tap water to measured plasma concentrations range from approximately 2–35 % suggesting additional exposure sources and pathways [32]. Along with inhalation of PFAS [52], dermal contact and ingestion of dust are also potentially important PFAS exposure pathways in homes and have been estimated to account for up to 25 % of the body burden for some PFAS [15]. House dust can act as an aggregate of PFAS in residential environments that is transferred from ambient sources or from indoor sources including materials and products in the residential environment [19,95]. Variability in PFAS concentrations in serum have been associated with demographic information and housing characteristics, which was thought to be driven by different types and uses of consumer products or materials in homes that influence PFAS levels in house dust [13,27,40]. Due to the prohibitive time, cost, and logistics of collecting house dust from people’s homes, previous studies investigating PFAS in house dust and potential drivers have mainly been limited in sample size and geographic extent [10,18,21,36,39,97]. Whether there is a relationship between PFAS in homes’ house dust and tap water has also not been investigated on a large in situ scale due to the lack of paired samples.

The American Healthy Homes Survey (AHHS) was a nationwide sampling campaign designed to assess the risks of residential hazards such as lead, mold, and pesticides with a focus on children [92]. It was implemented by the U.S. Department of Housing and Urban Development (HUD) and the U.S. EPA in 2005 (AHHS I) and again in 2017 (AHHS II) and involved collecting environmental samples, observational inspections of homes, and questionnaires from a nationally representative set of occupied primary residences. A subset of paired point-of-use tap water samples and house dust samples from the same AHHS II homes were subsequently available for PFAS analysis. Using ultra performance

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liquid chromatography-mass spectrometry (UPLC-MS), nine perfluoroalkyl carboxylic acids (PFCAs C4-C12) and seven perfluoroalkyl sulfonic acids (PFSAs C4-C10) were quantified. This study presents a large analysis of ionic PFAS in paired tap water-house dust samples from residential homes in the general U.S. population and is one of the first opportunities to characterize PFAS in house dust from the general population throughout the U.S. The paired samples were used to analyze potential relationships between PFAS in these media and to investigate indicators of PFAS in tap water and house dust using questionnaires and geospatial information in relation to known, suspected, or possible sources.

2. Methods

2.1. Sample collection

Details regarding AHHS II household recruitment and sample collection are described in U.S. HUD [92] and Bradham et al. [7]. Briefly, permanently occupied, non-institutional housing units in the United States where children may reside were targeted for a nationally representative participant recruitment. One hundred Primary Sampling Units (PSUs), clusters of housing located in Metropolitan Statistical Areas, single counties, or groups of counties, were randomly selected in AHHS I [92]. The 78 PSUs targeted for AHHS II were a subset of the 100 PSUs sampled in AHHS I [92]. Between March 2018 and June 2019, interviewers were dispatched to these locations for environmental sampling, visual inspections, and administration of questionnaires. The day before their interview, residents were instructed to collect composite tap water samples at intervals throughout the day in order to simulate exposures from daily tap water use. During the interview, resident vacuum bag or canister contents were collected. The vacuum dust and tap water samples were collected and stored based on procedures for the originally intended analyses for lead, mold, and pesticides. Vacuum dust samples were sieved to a final size fraction of < 150 μm. Interviewers administered questionnaires to household representatives that were recorded on a tablet PC. These questionnaires were tailored to ask about activities, behaviors, and housing characteristics indicative of risks to contaminants like lead-based paint. The house dust and tap water samples used in this study are from a subset of the AHHS II households that had both a sufficient amount of vacuum dust after sieving for laboratory PFAS analysis and a paired tap water sample from the same home (n=241). These 241 unique households were located in 141 cities throughout 36 U.S. states.

2.2. Analytical methods

2.2.1. Standards and reagents—Liquid chromatography-mass spectrometry (LC-MS) grade acetonitrile, ammonium hydroxide, and formic acid were purchased from Fisher Scientific (Fair Lawn, NJ) and methanol from Honeywell - Burdick & Jackson (Muskegon, MI). Ottawa Sand Standard (20–30 Mesh) and glacial acetic acid were purchased from Fisher Scientific. ACS reagent ammonium acetate, Supelclean™ ENVI-Carb™ SPE cartridges, and sodium acetate trihydrate were purchased from Sigma-Aldrich (St. Louis, MO). Waters Oasis WAX Plus Short Cartridges (225 mg sorbent, 60 μm) were purchased from Waters Corporation (Milford, MA). In-house deionized water was used. Native standard solution PFAC-MXC (2000 ng/mL) and mass-labelled PFC extraction standard

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solution MPFAC-C-ES (2000 ng/mL) were purchased from Wellington Laboratories (Ontario, Canada).

2.2.2. Tap water sample preparation—Residential drinking water samples collected as part of the AHHS II effort had been previously analyzed for trace metals and were acidified to 2 % nitric acid (w/w) and stored in 1 L HDPE bottles at 4 °C before being transferred for PFAS analysis during the summer of 2021. Samples were poured into a pre-weighed 1 L HDPE bottle to determine sample mass and calculate original sample amounts. The original 1 L HDPE bottle was rinsed with 10 mL methanol, shaken, and pooled with the weighed sample. Pooled water and methanol rinsate were spiked with mass-labelled internal standard (Wellington MPFAC-C-ES) to final concentration of 50 ng/L. Samples were concentrated via solid-phase extraction (SPE) following a modified version of EPA Method 533 [64]. Briefly, a 250 mL subaliquot of the water samples was concentrated via weak-anion exchange SPE using preequilibrated Waters Oasis WAX Plus Short Cartridge (225 mg sorbent, 60 μm) and a positive displacement pumping system. Offline SPEs were GL Sciences AQUA Loader Twin SPL 698 T and Chratec Sep-Pak Concentrator. After washing the SPE cartridges with 5 mL of pH 4 buffer and then 5 mL of methanol, samples were eluted from SPE using 5 mL of 0.1 % ammonia in methanol. Eluate was evaporated to less than 0.5 mL, but not to dryness, using a Caliper Life Sciences TurboVap LV nitrogen evaporator at 40 °C and 12 psi and reconstituted with methanol to a final volume of 0.5 mL. A 50 μL aliquot of the sample was combined with 150 μL of mobile phase A (95/5 deionized water/acetonitrile v/v + 2.5 mM ammonium acetate) in a clean autosampler vial and vortexed for 30 s prior to analysis. A matrix-matched calibration curve was prepared using in-house deionized water spiked with native standards and 2 % (w/w) nitric acid at the following calibration points (ng/L): 0, 0.1, 0.5, 1, 2.5, 5, 10, 25, 50, 100, 250. Extracted calibration curves were prepared in tandem with the AHHS samples, along with per-batch method blanks consisting of nitric acidified deionized water.

2.2.3. House dust sample preparation—After sample collection dust samples were stored at −20 °C until analyzed in the laboratory during the fall of 2021. Dust was processed using a method modified from a previously described method for PFAS dust analysis [21]. Dust was sieved through a shaker (Gilson Company, 1-Touch Vibratory Sieve Shaker SS-10) to < 150 μm, transferred to 20 mL glass scintillation vials and rotated in the x, y, and z planes for 1 minute to assure homogeneous mixing. After rotation, approximately 50 mg of material was removed to a 15 mL polypropylene centrifuge tube. To each centrifuge tube with dust, 2.5 mL of methanol containing 1.2 ng/mL of MPFAC-C-ES internal standard was added. Each tube was vortexed for 3 s followed by sonication for 30 min using a Branson 5510R-MT ultrasonicator. Then each tube was centrifuged at 3500 rpm for 10 min using a Thermo Scientific IEC CL31R Multispeed centrifuge. In the meantime, Supelclean™ ENVI-Carb™ Solid Phase Extraction (SPE) cartridges (bed wt. 250 mg, volume 3 mL) were primed with 5 mL methanol each. After centrifugation, samples were poured into SPE cartridges and eluate collected in polypropylene centrifuge tubes. Like the tap water samples described above, the eluate was concentrated under nitrogen flow to less than 0.5 mL but not to dryness, and then reconstituted with methanol to a final volume of 0.5 mL. A 100 μL aliquot of the sample was combined with 300 μL of mobile phase A (95/5 deionized water/

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acetonitrile v/v + 2.5 mM ammonium acetate) in a clean autosampler vial and vortexed for 30 s prior to analysis. Method blanks consisted of Ottawa Sand (20–30 mesh) grinded and sieved through a shaker. A matrix-matched calibration curve was prepared with native PFAS standards in Ottawa Sand spiked with mass-labelled internal standards at the following calibration points (ng/g): 0, 1, 5, 10, 25, 50, 100, 250, 500. The calibration curve was extracted exactly as described above.

2.2.4. Ultra performance liquid chromatography-mass spectrometry (UPLC-MS) analysis—

Extracted tap water and dust samples were analyzed via UPLC-MS with a Waters ACQUITY UPLC BEH C18, 130 Å, 1.7 μm, 2.1 mm×50 mm column (Milford, MA) at 55 °C on a Thermo Scientific Vanquish Horizon UPLC. A delay column Thermo Scientific Hypersil GOLD C18, 1.9 μm, 3 ×50 mm (Waltham, MA) between the static mixer and injector was used as part of standard practice. Chromatographic separation was accomplished with a 10 minute reversed-phase gradient using mobile phase A (95/5 deionized water/acetonitrile v/v + 2.5 mM ammonium acetate) and mobile phase B (5:95 deionized water/acetonitrile v/v + 2.5 mM ammonium acetate) (Table S1). Detection was carried out by a coupled Thermo Orbitrap Fusion Tribrid mass spectrometer in negative electrospray ionization mode. Data were collected at 50,000 resolving power with a preferred ion data-dependent acquisition for additional feature identification. Specific instrumental parameters for the LC separation and MS operation are listed in Table S2.

After data collection, chromatograms were processed, and peak areas were integrated in Thermo Scientific Xcalibur Quan Browser 4.3. Native standard peak areas were matched against internal standard peak areas according to Table S3 and response ratios were calculated. Statistical analyses were conducted using R, and the method reporting limit (MRL) and limit of detection (LOD) for each chemical and matrix were estimated using the *LCMRL* package [78]. MRL was defined by the lowest spiking concentration such that the 99 % confidence interval for the spike-recovery was within the range 50 % - 150 %, and LOD was defined with the 99 % confidence interval for the reported concentration being greater than 0. These MRL and LOD calculations were based on six repeat measurements of the extracted calibration curves across three different analysis batches over one month [78].

For quality assurance checks, continuous calibration standards were run every ten samples, alternating between 15 and 50 ng/L for water and 15 and 75 ng/g for dust. Accuracy was maintained with an average relative percentage difference < 20 %. Technical replicate analysis was conducted for a random 10 % subset of each matrix, finding an average replicate percent difference of < 1 %. Method blanks were repeatedly analyzed for tap water and dust to ensure clean background (one method blank per batch of tap water samples and every 60 samples for dust) and all reported analytes were consistently < MRL. The MRL and LOD for each chemical measured in tap water and/or house dust are listed in Table 1.

2.3. Statistical analysis

Detection frequencies for both media were calculated as the percent of samples in which PFAS concentrations were measured above the LOD. Concentrations that were measured above the LOD but below the MRL for each chemical were kept as reported by the

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laboratory in order to prevent bias from censored data estimation methods [23]. For chemicals in which the detection frequency was greater than 50 %, concentrations below that chemical's LOD were substituted with its LOD/√2 due to their being little bias found with this method for skewed distributions [23]. Chemicals with detection frequencies below 50 % were not used in subsequent statistical analyses due to the higher risk of statistical bias and for consistency with previous PFAS studies [21,23,61]. A ∑PFAS metric was calculated as the sum of the concentrations of PFAS with detections frequencies above 50 % for each medium. Summary statistics were calculated for ∑PFAS and the PFAS in each medium with greater than 50 % detection frequencies using R packages "vtable" and "EnvStats" [33,49]. Relationships between PFAS concentrations in paired tap water and house dust, as well as between PFAS in the same media, were investigated using Spearman rank correlations. All statistical analyses were performed in R statistical software version 4.2.1 [62].

2.3.1. Exposure estimation—Using the measured concentrations, estimates of daily intakes (ng/day) and serum concentrations attributed to the 4 PFAS (PFOA, PFOS, PFHxS, and PFHpA) that were detected in more than 50 % of both tap water and house dust samples were calculated to assess potential contributions to body burdens for simulated adults and children (ages 6 months to 10–11 years). Average daily intake (ADI) for water ingestion and dust ingestion were calculated using pathway-specific Eqs. (1) and (2), respectively, from Lorber & Egeghy [47].

$$ADI_{water\ ingestion} = C_{water} * V * AF \quad (1)$$

where C_{water} are the concentrations of PFAS measured in tap water samples in this study (ng/L), V is the mean (consumers-only) daily volume consumed (L/day) determined using the U.S. EPA's Exposure Factors Handbook [81], and AF is the gastrointestinal absorption fraction that was assumed to be 100 % for water and dust ingestion [26,72].

$$ADI_{dust\ ingestion} = C_{dust} * IR * AF \quad (2)$$

where C_{dust} are the concentrations of PFAS measured in house dust samples in this study (ng/g) and IR is the general population central tendency ingestion rate (g/day) determined using the Exposure Factors Handbook [80]. Body weights (bw) were also determined using the Exposure Factors Handbook [79]. A simple first-order pharmacokinetic model (3) that assumes steady-state was used to calculate estimate serum concentrations from the daily intakes for each pathway.

$$C_{serum} = DP/(kP * Vd) \quad (3)$$

where C_{serum} are concentrations of PFAS estimated in serum (ng/mL), DP is the daily absorbed dose (ng/kg-bw/day) calculated as the ADI divided by body weight, kP is the

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chemical-specific first-order elimination rate in the body (1/day), and Vd is the chemical-specific volume of distribution (mL/kg). All exposure factors, parameters, and equations used for exposure estimations in this study are listed in Table S4 [14,25]. Serum estimations (PFOA, PFOS, PFHxS) for adults were compared to measured serum concentrations from participants 21 years of age or older (n=2810) from NHANES cycle 2017–2020 [75]. Because NHANES does not report PFAS data for children under 12 years old, serum estimations (PFOA, PFOS, PFHxS) for children were compared to measured serum concentrations from participants 3–11 years of age (n=639) from NHANES cycle 2013–2014 [99].

2.3.2. Geospatial—Spatial variability in ∑PFAS for tap water and house dust in the AHHS homes was quantified using the global Moran's Index [53], calculated in ArcGIS Pro (version 3.1.0). This metric of spatial autocorrelation measures the tendency for similar measurements in each media to occur near each other. A distance band threshold of 500 km was used for inverse distance weighting in order to limit feature neighbors to those within similar regions of the country.

To facilitate an analysis of associations between ambient source proximity and media concentrations, geographic coordinates for a variety of potential PFAS sources were downloaded from USEPA's online data and mapping application called PFAS Analytic Tools [86]. These coordinates generally pointed to the center of each facility which did not account for specific areas or sources of contamination within larger facilities. Data sources, detailed descriptions of source categories, and the number of facilities within various regions around the U.S. can be found within the PFAS Analytic Tools site [86].

To quantify this spatial information on potential PFAS sources for statistical analysis, distances were calculated from the centroid of each zip code in which house dust and tap water samples were analyzed for PFAS to the coordinates of the nearest facility for 6 potential source categories – federal sites, national defense sites, EPA response locations, fire training sites, airports, and PFAS manufacturers/importers. Industries from EPA's Enforcement and Compliance History Online (ECHO) database [87] that could potentially use or manufacture PFAS and do not fall into the above categories, referred to as "Other ECHO industries" henceforth, were counted within a 5 km buffer around the center of the zip codes. This type of quantification was conducted for this source category instead of distance calculations due to the large number of potential PFAS-related industries in many areas. All spatial calculations and mapping were performed in ArcGIS Desktop (version 10.8.1). While zip code-level spatial information was available for AHHS II sample households for these spatial calculations, only city- and state-level information is reported here in order to protect participant privacy. Statistically significant associations (where $p < 0.05$) between log-transformed PFAS and ∑PFAS concentrations in household media and the quantified spatial information were determined using linear regressions in R statistical software. Where the p-value was greater than 0.05 but less than 0.10, the relationships are described as non-significant associations.

2.3.3. Questionnaire—Specific variables were selected from the AHHS questionnaire that pertained to previously observed associations between PFAS in house dust, tap water,

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or serum and demographic information, housing characteristics, or behaviors [28], *in review*; [6,9,12,16,36,37,44]. These variables included income, where respondents reported their household income as being above or below $35,000 per year, and the number of people living within the household. Variables were also selected where respondents reported whether any children under the age of 18 resided in the home, whether the home was owned or rented, and whether the home received their tap water from public water supplies such as central city or county sewer or from a private well. Whether tap water samples from public water supplies were sourced from groundwater or surface water is not known, and household locations are not necessarily reflective of public water intake locations. Responses where the participant was not sure, refused to answer, or answered "other" were not used in the analyses.

The responses for some questionnaire variables were combined and regrouped in order to produce a small number of response categories with larger numbers of households for more robust statistical analysis. Due to skewness towards white participants in this subset of homes, questionnaire responses regarding race were combined into two groups – participants that reported their race as white and participants that reported their race as non-white or biracial. While ethnicity was also asked in the AHHS questionnaire, not all participants that reported their race also reported their ethnicity. Because reported ethnicity was also highly skewed towards non-Hispanic participants (88 %), only self-reported race is used in this study. Many households in the study reported having one or two household members, so responses for the number of household members were grouped as either one or two members or three or more members. The time since the home had last been reportedly vacuumed was grouped into three categories – 1 month or less, 1 month to 3 months, or more than 3 months. Correlations between these categorical questionnaire variables were determined using chi-square tests.

For each response group, geometric mean and geometric standard deviations were calculated for ∑PFAS and PFAS in house dust and tap water. Not all households with PFAS measurement data provided responses for the entire questionnaire; the number of households used for statistical analyses varies based on which questionnaire variables are used. One-way ANOVAs were performed to determine associations between questionnaire responses and log-transformed concentrations of ∑PFAS and PFAS in house dust and tap water. Statistical significance was determined as $p < 0.05$, and associations with p-values greater than 0.05 but less than 0.10 were reported as non-significant associations.

3. Results and discussion

3.1. PFAS in tap water and house dust

Detection frequencies, the percent of samples in which the PFAS was measured above the limit of detection, for the paired house dust and tap water samples are shown in Table 1. Nine PFAS were detected in more than 50 % of the tap water samples – in descending order: PFBS, PFHpA, PFOS, PFOA, PFPeS, PFBA, PFHpS, PFHxA, and PFHxS – and nine PFAS were detected in more than 50 % of the house dust samples – in descending order: PFOA, PFNA, PFOS, PFHxS, PFDA, PFUnDA, PFDoA, PFHpA, and PFDS. Of the chemicals that were measured in both media, 4 were measured above 50 % in both house dust and tap

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water in these homes – PFOA, PFOS, PFHpA, and PFHxS. Longer-chain PFAS generally had greater detection frequencies in house dust, while shorter-chain PFAS generally had greater detection frequencies in tap water. All households in this study had at least one of the targeted PFAS detected in both their tap water and house dust.

The chemicals most frequently detected in the paired point-of-use tap water samples were PFBS (100 %), PFHpA (72 %), PFOS (70 %), and PFOA (65 %). Boone et al. [5] also found high detection rates of these chemicals in source waters and treated waters at drinking water treatment plants around the U.S., while another study also found that PFBS was the most frequently detected PFAS in point-of-use tap waters in the U.S. from both public water supplies and private wells [69]. PFHxS was not one of the more frequently detected chemicals in these homes in contrast to previous U.S. drinking water studies [5,69]. Detection frequencies for tap waters in this study were higher than those reported in another nationwide study of PFAS in tap water in the U.S., likely due to the higher limits of detection for these chemicals in the previous study's analytical method [69]. Where the previous study estimated a large proportion of left-censored concentrations with Bayesian modeling, this study was able to analyze measured data because of the high frequencies of detection.

Of the 9 PFAS with greater than 50 % detection frequencies in tap water, the highest median concentrations in the AHHS homes were PFBA (1.17 ng/L), PFOS (0.96 ng/L) and PFOA (0.83 ng/ < L) (Fig. 1, Table S5). These chemicals also had the highest median concentrations of the PFAS measured at drinking water treatment plants in the U.S., but the AHHS homes' tap water had lower median concentrations for each chemical [5]. The lowest median concentrations measured in tap water were PFPeS (0.1 ng/L), PFHpS (0.1 ng/L), and PFHxS (0.47 ng/L).

The chemicals most frequently detected above the limit of detection in the AHHS house dust samples were PFOA (97 %), PFNA (92 %), PFOS (87 %), and PFHxS (83 %). The chemicals least frequently detected above the limit of detection in house dust were PFPeS (23 %), PFPeA (10 %), and PFNS (8 %). These findings are consistent with many smaller studies previously reported in the literature [29].

Of the 9 PFAS with greater than 50 % detection frequencies in house dust in this study, the highest median PFAS concentrations were PFOS (9.48 ng/g), PFOA (8.68 ng/g), and PFDS (5.16 ng/g) (Fig. 1, Table S5). While PFOS was the chemical with the highest median concentration in house dust from homes in this study, previous studies from cities around the U.S. reported higher median concentrations of PFOS than was measured in this nationwide population [21,29,36,39,96]. Most of the same studies also measured higher median concentrations of PFOA in house dust than was observed in the AHHS II homes [21,29,39,96]. The house dust samples in these previous studies were collected years prior (2007–2014) to the AHHS II house dust samples (2018–2019), suggesting that PFOS and PFOA levels in U.S. house dust have declined during the last decade following the phase-out of these chemicals from consumer products. However, they remain the predominant PFAS species in these homes. The median concentration of PFDS from homes in this study was consistent with the median PFDS concentration reported in a previous U.S. study from

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homes in Wisconsin [39]. The lowest median concentrations in house dust were PFUnDA (1.02 ng/g), PFDoA (1.32 ng/g), and PFHxS (1.34 ng/g). While reported in only one or two studies, median concentrations of PFUnDA, PFDoA, and PFHxS in previous U.S. studies' house dust were higher than those found in house dust in this study [29,36,39].

3.2. PFAS concentration correlations

To test whether there was a relationship between PFAS concentrations in a home's tap water and house dust, Spearman correlations were performed for chemicals in which the detection frequency was greater than 50 % in both house dust and tap water and for cumulative PFAS concentrations. There was no correlation found for PFOA, PFOS, PFHpA, PFHxS, or ΣPFAS (ρ= −0.09, 0.01, −0.02, 0.18, and −0.02 respectively). While tap water may be used within households for activities like cleaning and showering, there does not appear to be a significant partitioning of these 4 PFAS from tap water into house dust in the homes in this study. There also does not appear to be a relationship between the cumulative amount of PFAS in a home's dust and tap water. Partitioning of PFAS between different media like water, air and soil varies by compound depending on its chain length, function group charge, and vapor pressure [1,35,57]. The influence of PFAS in tap water on PFAS in the home's dust should be explored further with additional compounds, particularly more volatile PFAS precursors. However, this result suggests that there are different types of sources driving ionic PFAS concentrations in the two media; PFAS in house dust is driven by within-home sources like consumer product use and materials in the home while PFAS in tap water is driven by ambient contamination sources in the local environment.

Spearman correlations were also used to assess relationships between PFAS within the same media. All PFAS were correlated positively within both tap water and house dust, indicating that exposure to one PFAS in these media increases the likelihood of being exposed to other PFAS in that media (Tables 2–3). All correlations within tap water and house dust were statistically significant ($p < 0.05$). The average ρ values across all PFAS in tap water and house dust were 0.65 and 0.63, respectively, which suggests similarly moderate extents of co-exposure to multiple PFAS. However, the perfluoroalkyl sulfonic acids (PFSAs) were more correlated in tap water (average ρ =0.80) than perfluoroalkyl carboxylic acids (PFCAs) (average ρ =0.59), while PFCAs were more correlated within house dust (average ρ =0.82) than PFSAs (average ρ =0.56). Within these classes of more correlated chemicals within each media, PFAS are generally more correlated with PFAS that are closer in chain length. PFHxA was on average more correlated with PFSAs in tap water than were the other PFCAs, and PFOS was on average more correlated with PFCAs in house dust than were the other PFSAs. PFBA was least correlated with other PFAS in tap water, and PFDS was least correlated with other PFAS in house dust. Overall these results showed different patterns of exposure within the two media. Tap water was more likely to contribute co-exposures to shorter-chain PFSAs, and PFHxA to a lesser extent, due to increased water solubility as chain length decreases [4]. Meanwhile, house dust was more likely to contribute co-exposures to longer-chain PFCAs, and PFOS to a lesser extent, which has also been observed in dust in other indoor environments in the U.S. [21,36].

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3.3. PFAS Composition

The composition of PFAS in tap water and house dust in 20 households with the highest ΣPFAS of each respective media are shown in Fig. 2. For homes with the highest ΣPFAS concentrations in tap water, the composition generally included more PFOS and PFOA (Fig. 2A). There are also higher proportions of PFHxS and PFHxA in tap water in these homes. PFOS, PFOA, and PFHxS have been identified in groundwater at sites contaminated from AFFF use, which has been shown to contribute to elevated levels of contamination in drinking water [1,3,31,51]. Homes in this group with larger proportions of PFHxA in tap water tended to have less PFHxS in tap water, and vice versa. Similar to the homes with the highest concentrations in tap water, AHHS homes with the highest ΣPFAS concentrations in house dust generally had larger proportions of PFOA, PFOS, PFDA, and PFHxS (Fig. 2B). While replacement PFAS were becoming more detectable in these households, previously phased-out legacy compounds like PFOA and PFOS still dominated the proportion of PFAS found in many of these homes' house dust.

3.4. Exposure estimates

Summary statistics for estimated serum concentrations for simulated adults and children, calculated using concentrations measured for PFAS with greater than 50 % detection rates in both tap water and house dust samples (PFOA, PFOS, PFHxS, and PFHpA), are shown in Table S6. For adults, the median estimated serum concentrations from tap water were higherv than those from house dust for all chemicals (Fig. 3). Where median estimates from tap water were 4 times higher than those from house dust for PFOA and PFOS, the median serum estimates for PFHxS and PFHpA for tap water were 15 and 17 times higher than those from house dust, respectively. Homes with the maximum concentrations of PFOA and PFHxS observed in house dust in AHHS homes accounted for higher serum estimates in adults than did homes with maximum concentrations of those chemicals in tap water. For adults with the highest levels of exposure of some PFAS in house dust, the signal in serum levels could potentially overshadow that from drinking water. Compared to the median measured serum concentrations from NHANES participants for PFOA, PFOS, and PFHxS (1.3, 3.1, and 1.1 ng/mL, respectively), the sum (tap water and house dust) of median estimated serum concentrations calculated in this study were lower (0.10, 0.10, and 0.16 ng/mL, respectively) (Table S6). NHANES did not report PFHpA concentrations.

For children, median serum estimates from tap water were also higher than those from house dust for PFHxS and PFHpA (Fig. 3). However, unlike adults, children had higher median serum estimates for PFOA and PFOS from house dust than from tap water. At the maximum levels of measured PFAS in house dust and tap water, children's estimated serum concentrations were 12 times higher from house dust than tap water for PFOA and PFHxS and 3 and 6 times higher from house dust for PFOS and PFHpA, respectively. Lifestage factors like hand-to-mouth behavior increases the ingestion rate of dust for children compared to adults, which has been shown to elevate body burdens of PFAS in children previously [15,37]. For NHANES children ages 3–11 years old, the median measured serum concentrations for PFOA, PFOS, and PFHxS were 1.9, 3.8, and 0.81 ng/mL, respectively [99]. The sum (tap water and house dust) of estimated median serum concentrations of PFOA, PFOS, and PFHxS for children calculated in this study (0.28, 0.28,

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and 0.28 ng/mL, respectively) were lower than the measured concentrations (Table S6). PFHpA was detected at a low frequency (19 %) in the NHANES samples [99].

While the two exposure pathways used to estimate serum concentrations in this study do not account for all of a person's potential exposures to PFAS over time, the trends demonstrated in the exposure estimation calculations, where children generally had higher levels of PFAS in serum than adults, are consistent with measured concentrations. The exposure estimates calculated from measured PFAS concentrations in tap water and house dust in this study highlight the importance of addressing potential sources of exposure to PFAS within homes in addition to ambient sources affecting communities' drinking water, particularly to reduce children's exposure to these chemicals.

3.5. Spatial distribution

The spatial distributions of ΣPFAS concentrations in tap water and house dust collected at AHHS homes throughout the U.S. are shown in Fig. 4. The highest ΣPFAS concentration in tap water was located in Pennsylvania, while the lowest ΣPFAS concentration was located in Alabama. The Moran's Index value for spatial autocorrelation of ΣPFAS in tap water in the AHHS homes was 0.61 with z-score and p-value of 6.86 and < 0.001, respectively, indicating that similar concentrations are likely to be clustered near each other. The distribution of cumulative PFAS in tap water in this study is consistent with that estimated by Smalling et al. [69], where higher concentrations were observed along the eastern and northern mid-west, and lower concentrations were observed in parts of the northwest and southwest.

Most locations where house dust samples were collected and measured have homes with varying ranges of ΣPFAS concentrations. The highest and lowest ΣPFAS concentrations in house dust were both located in Nebraska. The Moran's Index value for ΣPFAS in house dust in the AHHS homes was −0.02 with z-score and p-value of −0.27 and 0.79, respectively, indicating that similar concentrations were not clustered or dispersed in space more than if they were random. Large ranges in PFAS concentrations in house dust in the same geographical area have been observed previously [27,39,40].

The significant spatial autocorrelation of tap water concentrations provides evidence in agreement with previous literature that geographical factors, such as proximity to ambient contamination sources, are the main drivers of contamination in this exposure medium [31,67]. In contrast, the lack of spatial autocorrelation in house dust concentrations suggests that contamination of house dust is being primarily driven by within-home factors like consumer product use and materials in the home that are not influenced by geographical region.

3.6. Proximity to sources

A summary of the analyzed associations between proximity to potential ambient sources and PFAS with detection frequencies greater than 50 % in tap water and house dust are provided in Table S7; associations with cumulative PFAS concentrations in each medium are shown in Table 4. There were significant associations between concentrations for most PFAS in tap water with proximity to potential PFAS sources. Closer proximity from zip code centers

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to the nearest airports was significantly associated ($p < 0.05$) with higher concentrations of all PFAS and ΣPFAS measured in tap water. Similarly, being located closer to the nearest federal sites, national defense sites, and PFAS manufacturing facilities were also significantly associated ($p < 0.05$) with higher concentrations of all PFAS and ΣPFAS in tap water except for PFHpS, for which there was a non-significant association ($p=0.06$). Closer proximity to the nearest EPA response location was also significantly associated ($p < 0.05$) with higher concentrations of all PFAS and ΣPFAS in tap water, except for PFOA for which there was a non-significant association ($p=0.05$).

Federal sites, EPA response locations, and many national defense sites have already been identified as known or suspected sources of PFAS contamination in drinking water by several federal agencies [77,74,88,89]. The use of legacy AFFF formulas at airports and military aviation sites has been found to be an ongoing source of PFOA, PFOS, and PFHxS contamination in local environments [31,45,46,65,66,67]. Other PFAS in tap water associated with sources in this study have been observed after the oxidation of several AFFF formulas [30]. PFAS manufacturing facilities have been linked to contamination of drinking water and environmental media in several areas in the U.S., prompting lawsuits, legislation, and nationwide monitoring of these chemicals [2,90,56,94,70].

The number of other ECHO industries within a 5 km radius from the center of the zip codes was significantly associated ($p < 0.05$) with higher concentrations of all PFAS and ΣPFAS measured in tap water, except for PFHpA for which there was a non-significant association ($p=0.09$) and PFBA which in contrast had a significant negative association. Therefore, for all PFAS except PFBA, for which the opposite was observed, homes near more of these industry facilities had higher PFAS concentrations in their tap water. PFAS-related industrial sites were a significant predictor of PFAS detections and concentrations in contaminated public water supplies in the U.S. Hu et al., [31] and have been suggested as a presumptive contamination source [67]. The number of industry sources within 5 km buffers around sampling sites was positively correlated with developed land use in a previous study [69], so this metric may act as a proxy for urbanization and development that have been linked to increases in cumulative PFAS concentrations and PFAS detections in drinking water [48,69].

Closer distances to the fire training facilities were significantly associated ($p < 0.05$) with higher concentrations of relatively few PFAS chemicals (PFHpA, PFBS, and PFHxA) in tap water compared to other potential sources investigated here, and it was also not associated with cumulative PFAS concentrations. Compared to most of the other sources investigated here, fire training sites were relatively distant from AHHS zip codes (median distance = 38 km), which may contribute to their lack of associations with PFAS in tap water.

Compared to tap water, there were fewer significant associations between PFAS concentrations in house dust and proximity to potential ambient sources (Table 4, Table S7). Generally, homes closer to EPA response locations had higher PFAS concentrations in their house dust. Conversely, homes closer to PFAS manufacturing or importing facilities had lower PFAS concentrations in their house dust. However, PFAS manufacturing and importing facilities were the most distant source type from AHHS zip codes (median distance = 73 km).

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Closer proximity to the nearest federal sites was significantly associated ($p < 0.05$) with higher concentrations of PFNA, PFDA, PFUnDA, and PFDoA in house dust, and was non-significantly associated with PFHxS (p=0.06) and ΣPFAS (p=0.08). Closer distances to the nearest airports were also significantly associated ($p < 0.05$) with higher house dust concentrations of PFOS, PFDA, and ΣPFAS, and had non-significant associations with PFOA (p=0.08) and PFHpA (p=0.07) concentrations. Distance to national defense sites and fire training sites were not significantly associated with any PFAS or ΣPFAS concentrations in house dust, but higher PFDA concentrations were non-significantly associated (p=0.05) with closeness to the nearest national defense sites. The number of other ECHO industries within a 5 km radius of the zip code centers were also not significantly associated with house dust concentrations of any PFAS or ΣPFAS.

Whereas tap water is directly impacted by effluents from industrial facilities and percolating AFFF at use sites, house dust in homes proximal to PFAS ambient sources may be reflecting transport via air and airborne particulate through open doors or windows [22]. In addition, house dust may become contaminated through the track-in of soils, in which PFAS levels can be highly elevated near contaminated sites [60,68,8]. The likelihood of occupational exposures to PFAS may also be linked to the distance from residences to sources, where workers at these facilities could transport chemicals from the worksite to their house dust on their clothes, skin, shoes, and accessories [41].

3.7. Questionnaire

Associations between questionnaire information and PFAS in serum in the U.S. have been studied using national-scale datasets [9,12,16,37,55,59,73], but associations between questionnaire information and PFAS in tap water and house dust have been relatively less studied and within smaller populations [23,38,41,42]. The large sample size and nationwide scale of the PFAS measurements and questionnaire information in this study provides a unique opportunity to investigate potential drivers and indicators of PFAS in U.S. households for the general population. While there is not an obvious direct pathway between some questionnaire variables available for this study and PFAS in tap water, demographic information could be used to help identify populations with a propensity for drinking water contamination, such as fenceline and inner-city communities [34,44]. A summary of the analyzed associations between questionnaire information and all PFAS with detection frequencies greater than 50 % in tap water or house dust are provided in Tables S8–S9, respectively. Associations between questionnaire information and ∑PFAS are shown in Table 5.

A household in the general population's tap water source – either public water supply or private well – was significantly associated ($p < 0.05$) with tap water concentrations of all PFAS chemicals and ∑PFAS, except for PFHpS (Table S8). For all of these chemicals, households with public water supplies had higher geometric mean concentrations than households with wells. For cumulative PFAS concentrations, tap water from public water supplies was 69 % higher than tap water from private wells (Table 5). This finding was opposite to the trend found for lead in tap water in AHHS II homes, where households with private wells had higher concentrations of lead than those on public water supplies

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[7]. While another nationwide U.S. study found comparable concentrations of PFAS in tap water sourced from public water supplies and private wells, the higher detection limits in that study may have decreased sensitivity to these differences [69]. However, that same study and another focused on the eastern U.S. did find that drinking water from public water supplies were more likely to have at least one PFAS detected than drinking water from wells [48,69]. The home's self-reported water source was only significantly associated ($p < 0.05$) with house dust concentrations of PFNA and PFUnDA. In contrast to tap water, homes with private wells had higher geometric mean concentrations of these chemicals in dust than did homes with public water supplies. Understanding differences in PFAS levels and sources for homes with public water supply versus those with private wells will require additional measurement.

In this study, self-reported household income, race, and whether the home was owned or rented were not significantly associated with any PFAS or ∑PFAS concentrations in tap water in the general population in this study (Table S8). On the other hand, household income was significantly associated ($p < 0.05$) with house dust concentrations of PFOA, PFNA, PFHpA, PFDA, PFDoA, PFUnDA, and ∑PFAS (Table S3), and had a non-significant association ($p=0.05$) with PFOS concentrations. Higher geometric mean concentrations in house dust were observed for households that reported an annual income of more than $35,000 for all of these chemicals, and there was a 55 % difference in geometric means for cumulative PFAS concentrations (Table 5). Associations between income and PFAS in serum have also been reported where higher income also indicates higher serum PFAS [9,12,16,37,55,73]. Higher income households may have higher levels of PFAS in their house dust due to differences in use of products and furnishings.

Race was significantly associated ($p < 0.05$) with house dust concentrations of PFOA, PFOS, PFNA, PFHxS, PFDA, PFDoA, PFUnDA, and ∑PFAS, and had a non-significant association ($p=0.07$) with concentrations of PFHpA (Table S9). White participants had higher geometric mean concentrations of these chemicals in their house dust than non-white or biracial participants, and there was a 49 % difference in geometric means for cumulative PFAS concentrations (Table 5). However, the lack of ethnicity information with all households that reported race and the bias towards white participants limits conclusions that can be made about race and PFAS in house dust from this study. Race and ethnicity have been shown to be associated with PFAS serum concentrations previously, with white participants also having higher serum levels of some PFAS [59,6].

Whether the household was owned or rented was significantly associated ($p < 0.05$) with house dust concentrations of all PFAS and ∑PFAS except PFDS (Table S9). Homes that were owned had higher geometric mean concentrations of these chemicals in house dust than homes that were rented, with an 87 % difference in geometric means for ∑PFAS (Table 5). This association has not been reported previously, but there is a correlation between this variable and household income (Figure S1) that could help to explain the association. In multivariate regression models using income, race, and owner/renter status as predictor variables, there was not agreement in the statistical significance of these predictors for different PFAS and ∑PFAS. This suggests that the confounding effect of these correlated

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variables on PFAS concentrations in house dust is complex and difficult to tease apart from one another.

The number of household members was significantly associated ($p < 0.05$) with tap water concentrations of PFHpA, PFHpS, and PFPeS, while it had non-significant trends with PFOS, PFHxS, PFBA, and ∑PFAS (Table S8). Whether the household had children under age 18 was significantly associated with PFHpA and PFBS in tap water, and had non-significant trends with PFHxA, PFBA, and ∑PFAS (Table S8). Households with fewer members and without children under age 18 had higher geometric mean PFAS concentrations in tap water. In house dust, the number of household members and whether the household had children under age 18 were significantly associated ($p < 0.05$) with all PFAS and ∑PFAS, except for PFDS (for number of household members) and PFDoA (for children under 18) (Table S9). Like tap water, higher geometric mean concentrations in house dust were also observed in households with fewer members and without children under 18; the direction of this trend in other house dust studies has been inconsistent [21,36]. There was a 59 % and 48 % difference in geometric means for ∑PFAS concentrations in house dust for homes with information on the number of household members and whether children under 18 lived in the home, respectively (Table 5). More frequent cleaning has been observed in households with more people, which could help to explain these trends in house dust [54]. Serum PFAS concentrations have also been shown to increase with decreasing number of household members [16]. However, questionnaire responses for the number of people in the household and income were correlated, so this association may again be primarily due to consumer product use influenced by income (Figure S1).

The amount of time since the household reported they had last vacuumed was significantly associated ($p < 0.05$) with house dust concentrations of PFOA, PFNA, PFHpA, PFDS, and PFUnDA, while PFOS and ∑PFAS had non-significant associations ($p=0.06$). Homes reporting they had last vacuumed more than 3 months prior had higher geometric mean concentrations than those who reported more recent vacuuming. Increased frequency of cleaning in the home has been associated with lower levels of PFAS in dust in previous studies [100,50]. Questions regarding flooring types and the types of furnishings and building materials present in the homes were not included in the questionnaire, but these could provide useful insight when included in future studies.

Home age had a non-significant association ($p=0.09$) with concentrations of PFOA in tap water but was not significantly associated with tap water concentrations of any PFAS or ∑PFAS. The age of the home was significantly associated ($p < 0.05$) with house dust concentrations of PFOS, PFNA, PFDS, PFUnDA, and ∑PFAS, and had a non-significant association ($p=0.08$) with PFDA concentrations. Homes built in 1980 or before had higher geometric mean concentrations of these chemicals in house dust than homes built after 1980. There was a 52 % difference in geometric means for ∑PFAS (Table 5). This finding was inconsistent with several previous studies that found lower levels of PFAS in dust in older homes [36,40,100]. However, geographical clustering of homes built in similar age ranges may be closer to a contaminated site, which could distort the previously observed trend in different locations [16].

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3.8. Limitations

Because the tap water and dust samples were initially collected to study indoor contaminants other than PFAS, sampling measures to prevent PFAS contamination were not implemented and field blanks that would be required for optimal quality control were not available for this study. As a result, there is uncertainty about the potential for background levels of sample contamination. Additionally, a storage stability test was not conducted to evaluate the samples before the UPLC-MS analyses. However, the containers used to store samples have previously been characterized for PFAS and were not expected to significantly impact the targeted legacy PFAS measured here [93,91].

The analytical methods used in this study measure a smaller suite of ionic PFAS in tap water and house dust, which may miss potential contributions to exposure from volatile PFAS or precursors. Future analyses will be conducted to identify any additional compounds that could be present and measured in these samples. The addition of paired measurements of PFAS in indoor air in future studies would also aid investigations into the full range of exposure pathways and PFAS compounds in homes.

The exposure calculations used in this study to estimate serum concentrations were intended to provide first-order insight on the potential implications of the PFAS measurements in tap water and house dust for adults and children that could have been living in the homes. The steady-state pharmacokinetic model used to estimate serum concentrations here may not be as appropriate for growing children as it is for adults, and the use of dynamic models could better account for exposures over time for PFAS with longer half-lives [17]. Because the ages of children that could have been living in the AHHS II homes were unknown, a default methodology using the averages of exposure factors from several childhood age ranges was used to calculate a generalized childhood exposure estimation. However, behaviors and contributions from different exposure pathways vary widely throughout childhood years so age-specific analyses could be done for children in the future.

Ideally, geographic proximity analyses would be conducted from individual home locations instead of from approximate locations using zip codes. However, in order to preserve participant confidentiality the exact locations of AHHS II homes were not available for this analysis. While the AHHS II study is nationally representative for the United States, there are gaps in geographic coverage among the states for homes that had paired tap water and house dust samples available for PFAS analyses conducted in this study. In addition, proximity analyses from potential PFAS sources generally pointed to the center of facilities but could not account for specific areas or sources of PFAS within facilities with larger areas. Finer resolution spatial analyses could improve the robustness of results in future studies where confidentiality is not a concern.

4. Conclusion

This study analyzed a large dataset of PFAS measurements in paired tap water and house dust samples. Higher detection rates in the paired tap water samples compared to previous studies helped to better characterize PFAS contamination in drinking water in the U.S. at the

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point of use. This study also presents one of the first nationwide datasets of PFAS in house dust across the U.S. general population.

Every PFAS that was measured in each medium was detected in at least one U.S. home. Generally, longer-chain PFAS had greater detection frequencies in house dust and shorter-chain PFAS had greater detection frequencies in tap water. The highest median concentrations in house dust were PFOA and PFOS, while the highest median concentrations in tap water were PFBA and PFOS. The composition of PFAS in tap water and house dust in homes with the highest cumulative PFAS levels showed that few households had the highest levels in both media, but there were substantial proportions of PFOA and PFOS in both tap water and house dust in the homes with the highest cumulative PFAS levels. Generally, PFOA and PFOS concentrations in house dust appeared to have declined compared to earlier studies.

Estimated serum concentrations for adults that were attributable to PFAS levels measured in tap water in this study were higher than those attributable to house dust for the same chemicals. However, some chemicals in house dust contributed to higher estimated serum concentrations than from tap water in children. Within the same media, concentrations of PFAS were positively correlated with each other. PFSAs of similar chain length were most correlated in tap water, while PFCAs of similar chain length were most correlated within house dust. These findings highlight the need for future studies and decision makers to consider mixtures and co-exposures of these chemicals. While there was not a direct relationship between PFAS concentrations in house dust and tap water in this study, this potential source of indoor contamination should be investigated further for more volatile compounds, particularly PFAS precursors. In addition, this result indicates that there are generally different sources of ionic PFAS concentrations for these two exposure media.

Similar geographical patterns in cumulative PFAS levels in drinking water were observed in this study, as were in previous studies. However, a larger dataset of unpaired tap water PFAS measurements will aid in this characterization throughout the U.S. in future work. More variability in cumulative PFAS concentrations within similar locations was observed in house dust than in tap water. Tap water contamination was more likely to be clustered geographically around ambient sources of PFAS, while house dust contamination was more likely to be individualized at the household level. Questionnaire responses regarding demographic information and housing characteristics were more often associated with PFAS concentrations in house dust than tap water. Overall the results from this study show that PFAS in house dust was mainly driven by within-home sources like consumer product use and materials in the home, while PFAS in tap water was mainly driven by ambient contamination sources in the local environment.

The findings from this study highlight differences in residential exposures and potential sources of PFAS contamination in tap water and house dust within homes in the general population in the U.S. These can aid in the identification and understanding of sources of PFAS contamination in impacted communities and households to facilitate future mitigation strategies and potentially inform regulatory decisions.

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Supplementary Material

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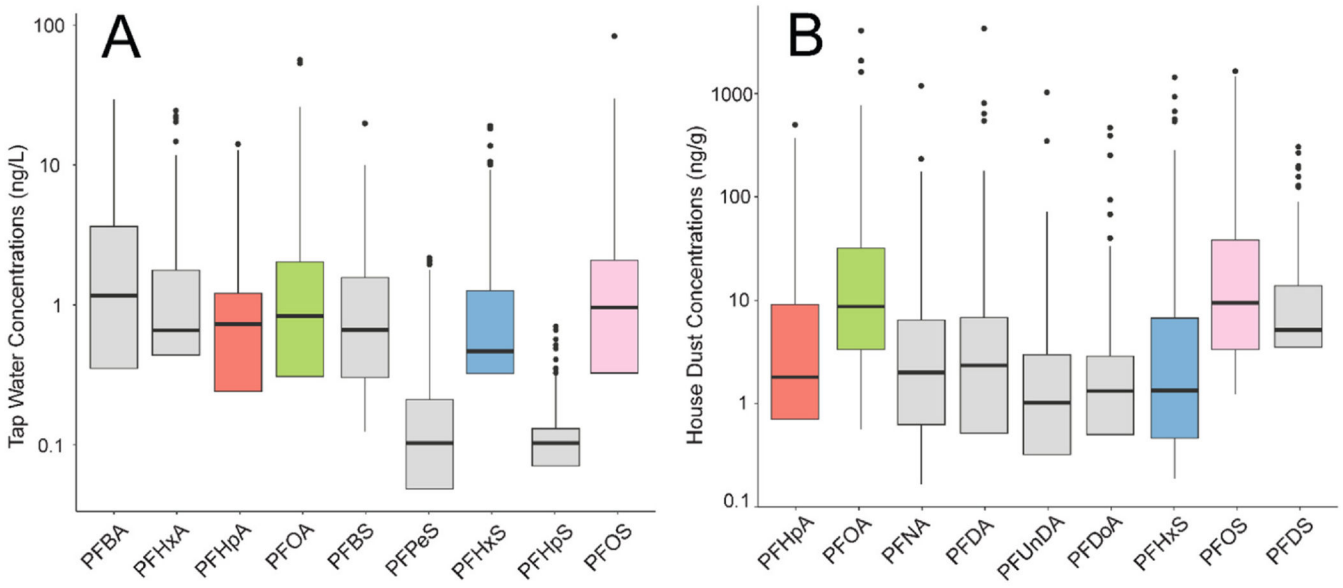


Fig. 1.

Boxplots showing distributions of PFAS concentrations in A) tap water (ng/L) and B) house dust (ng/g) for chemicals with detection frequencies greater than 50 %. Y-axes shown on log-scales. Boxplots shown in color correspond to chemicals measured (or detected > 50 %) in both media, whereas boxplots shown in gray indicate that the chemical was only measured (or detected > 50 %) in one of the media.

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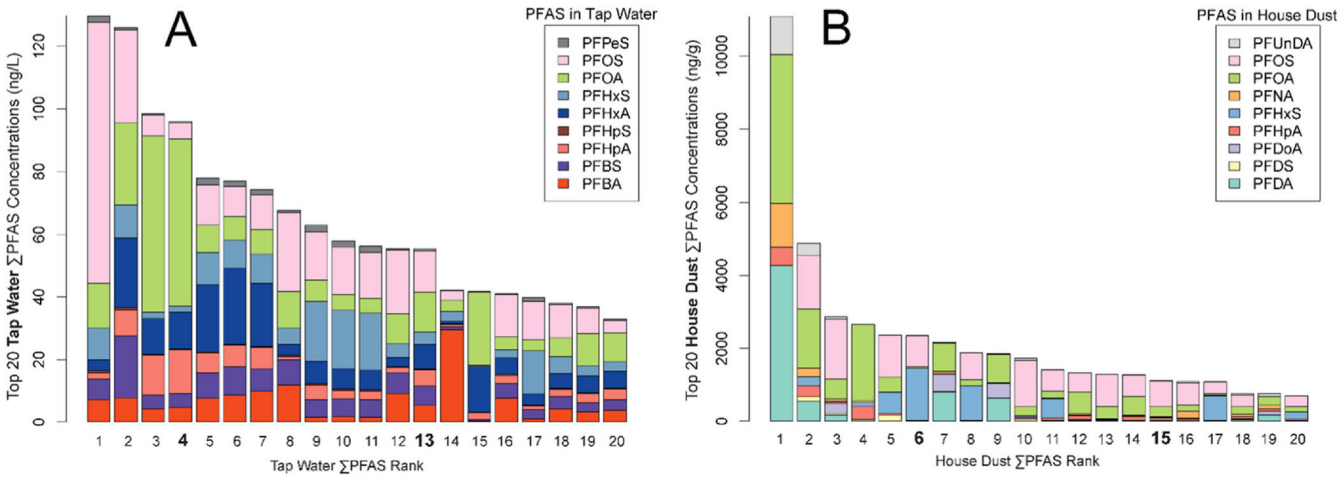


Fig. 2.

Composition of PFAS in homes with the 20 highest ∑PFAS concentrations in A) tap water (ng/L) and B) house dust (ng/g). Rank numbers for each media do not indicate measurements from the same homes except for two homes (#tap water-house dust) that are bolded (#13-6) and (#4–15).

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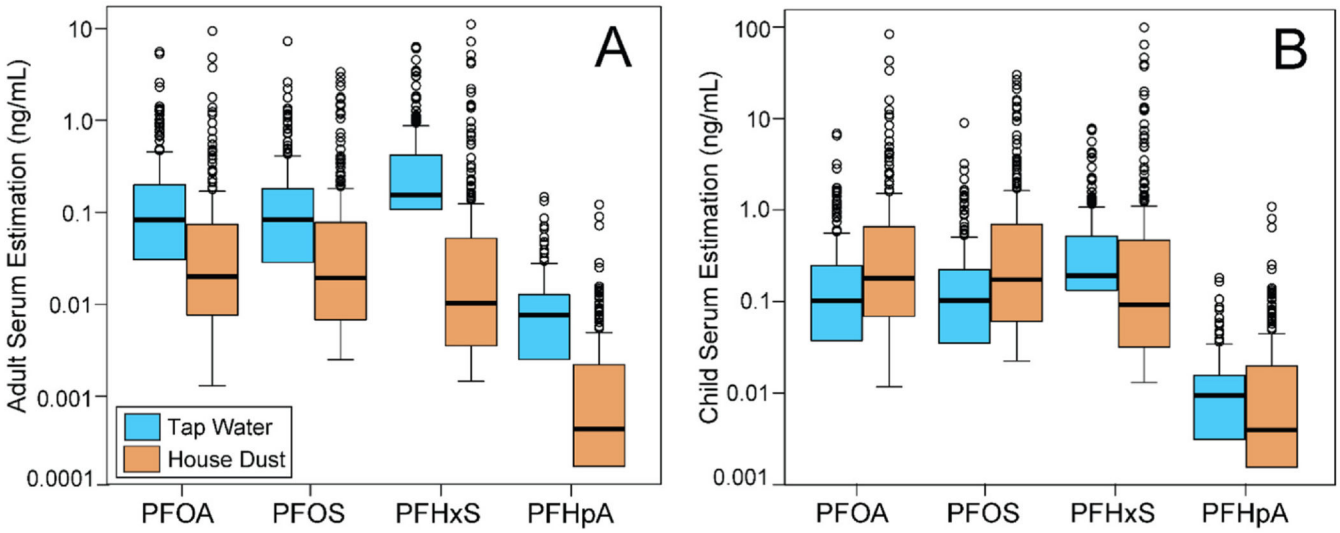


Fig. 3.

Estimated serum concentrations (ng/mL) for A) adults and B) children (ages 6 months to 10–11 years) due to exposure to PFAS through ingestion of tap water (blue) and house dust (brown). Daily ingestion intakes were calculated using concentrations of PFAS measured above 50 % detection frequencies in both media in the subset of AHHS II homes with paired samples.

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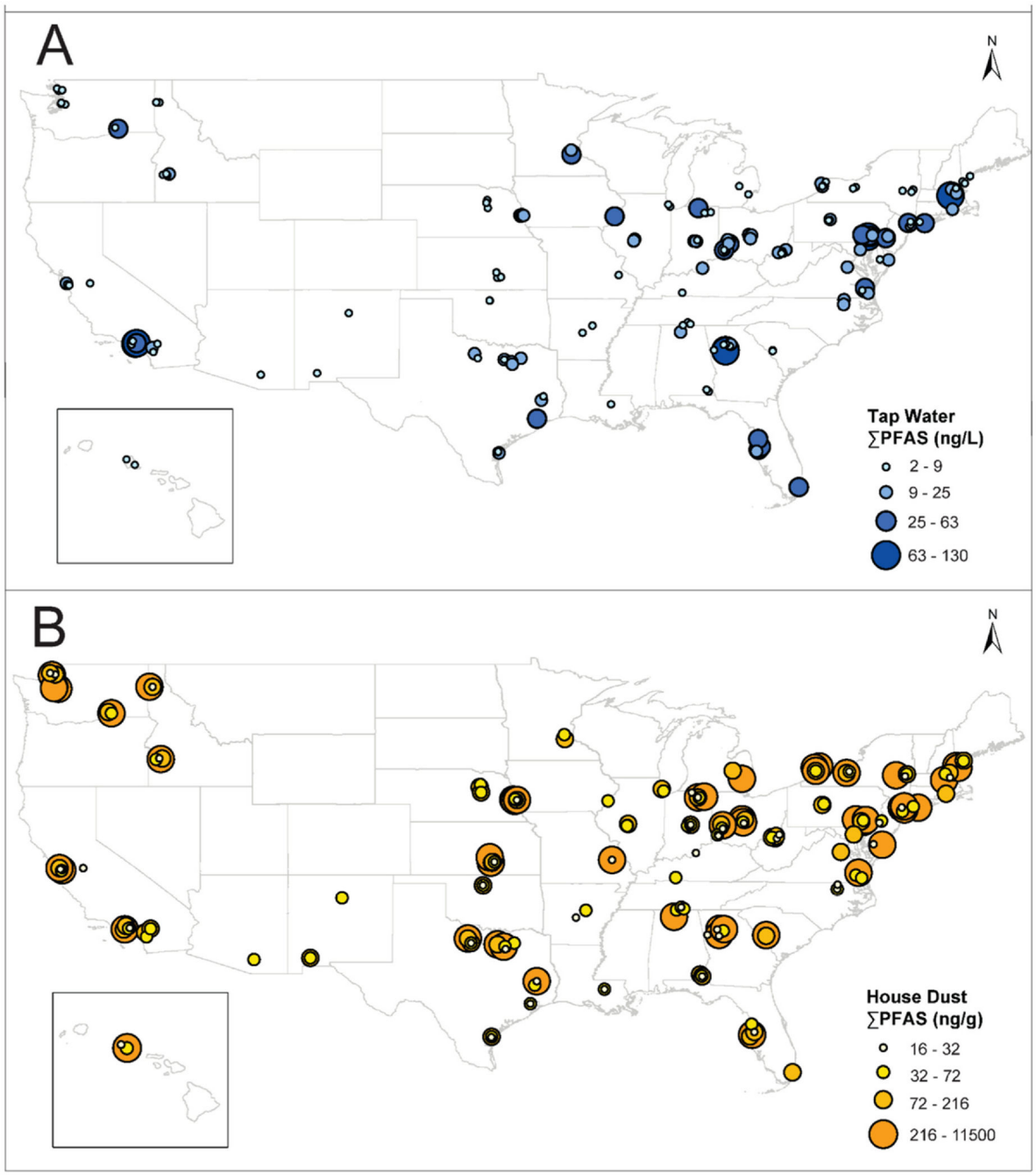


Fig. 4.

Maps showing spatial distributions of ∑PFAS concentrations in A) tap water (ng/L) and B) house dust (ng/g) in the subset of AHHS II homes with paired samples. Hawaii is shown in map insets.

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Table 1

Summary of targeted perfluoroalkyl carboxylic acids (PFCAs) and perfluoroalkyl sulfonic acids (PFSAs) measured in tap water and house dust, including chemical abbreviation, carbon chain length, detection frequency (%), method reporting limit (MRL), and limit of detection (LOD) for each chemical.

			Chain length	Tap Water			House Dust		
	Chemical	**Abbrev.**		**% > LOD**	**MRL (ng/L)**	**LOD (ng/L)**	**% > LOD**	**MRL (ng/g)**	**LOD (ng/g)**
PFCAs	Perfluorobutanoic acid	PFBA	4	58.92	2.76	0.50	37.34	7.05	3.08
	Perfluoropentanoic acid	PFPeA	5	46.47	2.79	1.73	10.37	3.56	2.14
	Perfluorohexanoic acid	PFHxA	6	51.87	0.89	0.62	34.85	6.49	3.79
	Perfluoroheptanoic acid	PFHpA	7	71.78	0.78	0.34	59.34	2.04	1.00
	Perfluorooctanoic acid	PFOA	8	65.15	0.75	0.44	96.68	1.86	0.80
	Perfluorononanoic acid	PFNA	9	25.31	0.78	0.46	91.70	0.50	0.23
	Perfluorodecanoic acid	PFDA	10	6.64	0.54	0.37	74.69	1.90	0.73
	Perfluoroundecanoic acid	PFUnDA	11	-	-	-	73.03	0.83	0.46
	Perfluoroundecanoic acid	PFDoA	12	-	-	-	67.63	1.26	0.71
PFSAs	Perfluorobutanesulfonate	PFBS	4	100.00	0.33	0.11	39.00	2.44	1.20
	Perfluoropentanesulfonate	PFPeS	5	63.49	0.43	0.07	22.82	0.58	0.35
	Perfluorohexanesulfonate	PFHxS	6	50.21	0.66	0.46	82.99	0.82	0.27
	Perfluoroheptanesulfonate	PFHpS	7	53.94	0.35	0.10	30.71	0.58	0.30
	perfluorooctanesulfonate	PFOS	8	70.12	1.09	0.46	87.14	0.57	1.74
	Perfluorononanesulfonate	PFNS	9	0.83	0.72	0.43	8.30	1.90	1.13
	Perfluorodecanesulfonate	PFDS	10	-	-	-	51.87	22.21	4.96

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Table 2

Spearman rank correlations (ρ) for pairwise co-correlations of PFAS concentrations in tap water samples. Darker reds indicate increased correlation between chemicals, while lighter reds and whites indicate little or no correlation.

	PFBA	PFHxA	PFHpA	PFOA	PFBS	PFPeS	PFHxS	PFHpS	PFOS
PFBA	-	0.46	0.52	0.33	0.52	0.42	0.39	0.41	0.47
PFHxA	0.46	-	0.75	0.79	0.79	0.68	0.66	0.65	0.70
PFHpA	0.52	0.75	-	0.69	0.65	0.54	0.51	0.52	0.63
PFOA	0.33	0.79	0.69	-	0.73	0.66	0.64	0.63	0.73
PFBS	0.52	0.79	0.65	0.73	-	0.82	0.79	0.69	0.80
PFPeS	0.42	0.68	0.54	0.66	0.82	-	0.94	0.79	0.82
PFHxS	0.39	0.66	0.51	0.64	0.79	0.94	-	0.79	0.82
PFHpS	0.41	0.65	0.52	0.63	0.69	0.79	0.79	-	0.80
PFOS	0.47	0.70	0.63	0.73	0.80	0.82	0.82	0.80	-

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Table 3

Spearman rank correlations (ρ) for pairwise co-correlations of PFAS concentrations in house dust samples. Darker reds indicate higher correlation between chemicals, while lighter reds and whites indicate lower correlation.

	PFHpA	PFOA	PFNA	PFDA	PFUnDA	PFDoA	PFHxS	PFOS	PFDS
PFHpA	-	0.89	0.81	0.77	0.77	0.67	0.59	0.70	0.45
PFOA	0.89	-	0.84	0.83	0.79	0.73	0.66	0.80	0.47
PFNA	0.81	0.84	-	0.88	0.93	0.75	0.52	0.58	0.35
PFDA	0.77	0.83	0.88	-	0.90	0.87	0.47	0.55	0.30
PFUnDA	0.77	0.79	0.93	0.90	-	0.80	0.48	0.55	0.31
PFDoA	0.67	0.73	0.75	0.87	0.80	-	0.41	0.51	0.25
PFHxS	0.59	0.66	0.52	0.47	0.48	0.41	-	0.81	0.41
PFOS	0.70	0.80	0.58	0.55	0.55	0.51	0.81	-	0.47
PFDS	0.45	0.47	0.35	0.30	0.31	0.25	0.41	0.47	-

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Table 4

Coefficient estimates and p-values for linear regressions between distance (km) from AHHS zip codes to nearest facilities and log-transformed ΣPFAS concentrations for tap water and house dust. The model for “Other ECHO Industries” uses a count of facilities within 5 km from the AHHS zip code instead of distance. Bolded values show statistically significant ($p < 0.05$) trends. Negative estimate values indicate that ΣPFAS concentrations increased as distance to the source decreased, and vice versa. Positive estimate values for Other ECHO Industries indicate that ΣPFAS concentrations increased when homes were located near more industry facilities.

Ambient Sources	Tap Water ΣPFAS		House Dust ΣPFAS	
	Estimate	p-value	Estimate	p-value
Federal Sites	**−0.0063**	0.0009	−0.0053	0.0811
National Defense	**−0.0076**	0.0003	−0.0039	0.2492
EPA Response Locations	**−0.0070**	0.0014	**−0.0081**	0.0214
Fire Training	−0.0019	0.1371	−0.0025	0.2338
Airports	**−0.0127**	< 0.0001	**−0.0108**	0.0215
PFAS Manufacture/Import	**−3.89E-04**	< 0.0001	**0.0006**	0.0002
Other ECHO Industries[a]	**0.0077**	0.0001	0.0022	0.4809

[a]Count of industries within 5 km instead of distance.

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Table 5

Geometric mean (geometric standard deviation) ΣPFAS concentrations in tap water (ng/L) and house dust (ng/g) for questionnaire responses from the subset of AHHS II homes. Statistically significant associations ($p < 0.05$) from ANOVAs between questionnaire variables and log-transformed ΣPFAS concentrations are marked with an asterisk (*), and non-significant associations ($0.05 < p < 0.10$) are marked with an apostrophe (‘). N denotes the number of responses for each category.

Survey Question	n	Tap Water ΣPFAS	House Dust ΣPFAS
Annual Household Income			
Less than $35 K	93	8.05 (2.56)	46.53 (3.61)*
More than $35 K	139	7.97 (2.46)	82.09 (4.92)
Race			
White	183	8.08 (2.58)	72.48 (4.30)*
Non-White or Biracial	58	8.73 (2.55)	43.82 (5.08)
Number of Household Members			
1 or 2	141	9.09 (2.62)’	82.76 (4.76)*
3 or more	100	7.16 (2.46)	44.90 (3.93)
Children Under Age 18 in Home			
Yes	87	7.16 (2.49)’	47.00 (4.89)*
No	154	8.91 (2.59)	76.59 (4.24)
Own or Rent Home			
Own	163	7.84 (2.60)	86.71 (4.76)*
Rent	78	9.13 (2.48)	34.28 (3.29)
Home Age			
Built in 1980 or Before	105	7.55 (2.49)	94.04 (4.58)*
Built After 1980	55	9.00 (2.94)	52.64 (4.65)
Time Since Last Vacuumed			
1 month or less	148	-	58.23 (4.31)’
1–3 months	43	-	53.67 (4.91)
> 3 months	26	-	119.60 (4.78)
Tap Water Source			
Public water supply	216	8.77 (2.49)*	62.33 (4.44)
Private well	21	4.28 (2.96)	104.28 (5.14)